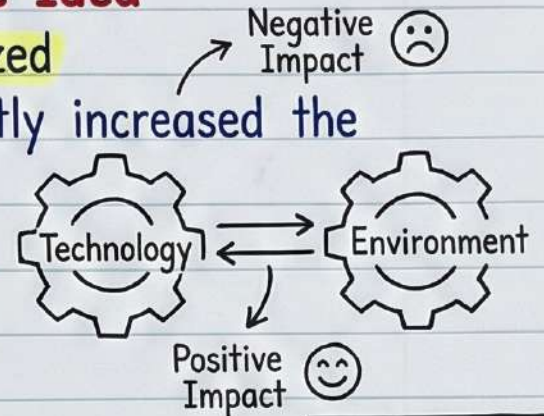


IMPACT OF COMPUTER TECHNOLOGY ON ENVIRONMENT

(SOC 2101 – Society, Environment and Engineering Ethics)
Lec 05-environmental-effect

1. Technology and Environment: Basic Idea

- Technological evolution has revolutionized human life, but it has also significantly increased the consumption of natural resources.
- Technology and environment are interlinked, and every technological invention has a direct or indirect impact on the environment.
- These impacts can be both negative and positive.



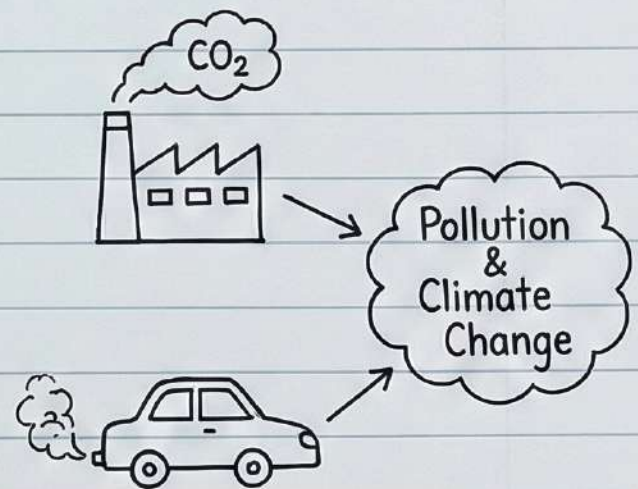
Exam Tip:
Don't forget
positive impacts!

2. Negative Impacts of Technology on Environment (Cons)

(a) Carbon Emission and Pollution

- Excessive CO₂ emissions from:
 - factories
 - power stations
 - vehicles
 - fossil fuel burning

Causes: • air pollution
• water pollution
• climate change



(b) Massive Energy Consumption

- Manufacturing electronic devices requires huge amounts of energy.
- Leads to a large carbon footprint throughout the product lifecycle.

(c) E-Waste (Electronic Waste)

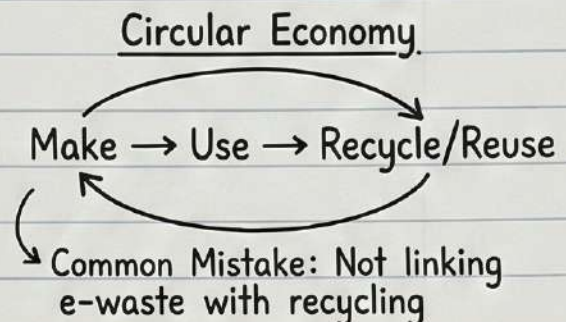
- Damaged or unused electronics often:

- end up in landfills
- are burned in open dumps

Linear Economy.
Take → Make → Dispose (Landfill) ✗

- Causes:

- soil contamination
- water pollution
- human health risks



Important analysis-based data:

- 20–50 million tonnes of e-waste generated worldwide each year.
- In the UK, around 1 million tonnes of electrical waste is disposed annually.
- EU regulations require collection of 45 tonnes of e-waste per 100 tonnes of electronics sold (previous 3 years).

(d) Depletion of Natural Resources

- Technology requires:
 - fossil fuels
 - rare earth metals
 - water
- Increased tech production leads to faster depletion of natural resources.

3. Positive Impacts of Technology on Environment (Pros)

(a) Renewable Energy

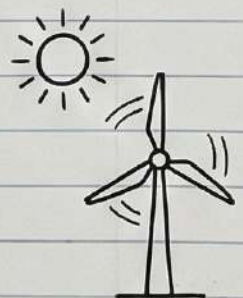
- Technology enables use of clean energy sources:

- solar
- wind
- water
- geothermal

- Devices such as:

- solar panels
- wind turbines

- Reduce dependence on fossil fuels and lower emissions.





(b) Smart Technology

- Smart homes and IoT-based systems improve energy efficiency.
- Examples:
 - ↳ intelligent lighting (lights only when needed)
 - ↳ smart thermostats (reduce energy waste)
- Leads to reduced electricity consumption.

(c) Electric Vehicles



- Powered by rechargeable batteries instead of fossil fuels.
- Reduce:
 - ↳ air pollution
 - ↳ greenhouse gas emissions
- Production increased since 2008.
- Expected to be cheaper than gasoline cars in future years.

(d) Reduced Resource Use Through Digitalization

- Email and digital work reduce paper usage.
- Remote work:
 - ↳ reduces commuting
 - ↳ saves fuel
- Video conferencing reduces travel-related emissions.

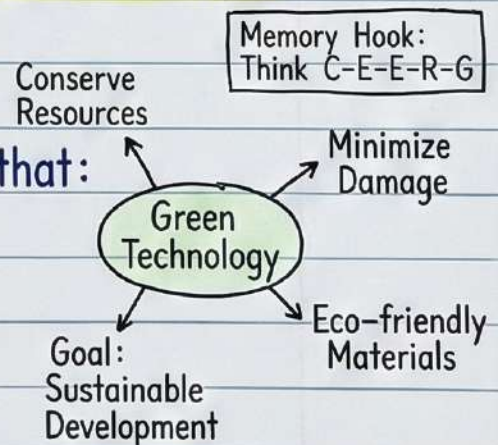
4. Green Technology



- Green technology refers to technologies that:

- ↳ conserve natural resources
- ↳ minimize environmental damage
- ↳ use eco-friendly materials and methods

- Goal: sustainable development with minimal environmental harm.



5. Case Study 1: Environmental Cost of Device Manufacturing

↳ Context

- Modern devices have short lifespans (2–3 years).
- Manufacturing a computer requires:
 - 3× its weight in fossil fuels
 - 630g fossil fuels for 1g silicon wafer
 - 1.5kg fuel + chemicals for 2g microchip
 - ~30kg clean water
 - ~1.5 tons of water per PC
 - ~22kg harmful chemicals (e.g., lead)

↳ Environmental Impact

- High water consumption
- Toxic chemical pollution
- Resource depletion
- Long-term ecosystem damage

↳ Ethical Measures for Companies

- Use sustainable materials
- Increase device lifespan
- Reduce toxic substances
- Improve manufacturing efficiency
- Invest in green production methods

6. Case Study 2: Environmental Impact of E-Waste

↳ Context

- Global e-waste value: \$62.5 billion per year
- Improper disposal causes:
 - environmental pollution
 - human health risks

↳ Ethical & Practical Solutions

- Promote proper recycling systems
- Encourage reuse and refurbishment
- Recover valuable materials:
 - gold
 - aluminium
 - rare metals
- Support specialized recycling companies

Exam Tip: Don't skip case study analysis!



7. Role of Recycling in Environmental Protection

- Recycling significantly reduces energy use:
 - Aluminium → 95% energy saving
 - Plastics → 70%
 - Steel → 60%
 - Paper → 40%
 - Glass → 30%
- Also reduces:
 - air pollution
 - water contamination
 - greenhouse gas emissions

8. Common Exam Mistakes (Avoid These)

- Writing only 'technology is bad'
- Ignoring positive impacts
- Forgetting figures from image slides
- Not linking e-waste with recycling
- Skipping case study analysis

9. Keywords & Memory Hooks

- Think C-E-E-R-G
 - Carbon emission
 - Energy consumption
 - E-waste
 - Recycling
 - Green technology

10. Likely Question Patterns

- Discuss positive and negative impacts of technology on environment
- Explain e-waste and its environmental effects
- Analyze a case related to device manufacturing or recycling
- How can green technology reduce environmental damage?



Good Luck!
😊